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Fuel remaining amount measuring apparatus, boat, and movable body

Abstract

A fuel remaining amount measuring apparatus for a boat includes a fuel tank containing fuel, a load sensor to measure a weight of the fuel tank, and a corrector to receive a measured value of the weight of the fuel tank detected by the load sensor. The corrector is operable to correct an error due to at least one of a movement or a tilt of the boat.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS
(1) The present application claims priority to Japanese Patent Application No. 2022-120184, filed on Jul. 28, 2022. The contents of this application are hereby incorporated herein by reference in their entirety.
BACKGROUND OF THE INVENTION
1. Field of the Invention
(2) The technology disclosed in the present specification relates to a fuel remaining amount measuring apparatus, a boat, and a movable body.
2. Description of the Related Art
(3) A boat is provided with a propulsion device. The propulsion device includes an internal combustion engine, for instance, and uses fuel contained in a fuel tank to generate a propulsion force.
(4) Generally, a remaining amount of fuel in a fuel tank is measured by using a float to detect a liquid level in the fuel tank. Depending on the shape of the fuel tank, however, the lowering of the liquid level per unit volume of decreasing fuel is not necessarily uniform in amount. In addition,

the liquid level in the fuel tank may vary with the movement or tilt of the boat. Consequently, if a fuel remaining amount measuring method based on the detection of the liquid level is used, the measurement accuracy may be reduced.
(5) In order to improve the accuracy of fuel remaining amount measurement, it is conventionally proposed to use a load sensor to measure the weight of the fuel tank so as to measure the remaining amount of the fuel (see JP 2004-37085A, for instance).
(6) A measured value of the weight of the fuel tank detected by using the load sensor may vary with the movement or tilt of the boat. Therefore, a conventional fuel remaining amount measuring method where the load sensor is used to measure the weight of the fuel tank still leaves room for improvement in the measurement accuracy. This problem is not unique to the fuel remaining amount measurement in the fuel tank provided on the boat but is also a problem for the fuel remaining amount measurement in a fuel tank provided on a movable body such as a vehicle or an aircraft.
SUMMARY OF THE INVENTION
(7) Preferred embodiments of the present invention provide solutions to the above problem.
(8) Preferred embodiments of the present invention are realized in the following aspects, for instance.
(9) A fuel remaining amount measuring apparatus for a boat according to a preferred embodiment of the present invention includes a fuel tank containing fuel, a load sensor to measure a weight of the fuel tank, and a corrector to receive a measured value of the weight of the fuel tank detected by the load sensor, and to correct an error due to at least one of a movement or a tilt of the boat.
(10) In this fuel remaining amount measuring apparatus, the detection of the liquid level in the fuel tank with a float is not utilized, but the fuel remaining amount measurement in which an error due to the movement or the tilt of the boat is corrected is performed, which improves the accuracy of the fuel remaining amount measurement.
(11) Another fuel remaining amount measuring apparatus for a movable body according to a preferred embodiment of the present invention includes a fuel tank containing fuel, a load sensor to measure a weight of the fuel tank, and a corrector to receive a measured value of the weight of the fuel tank detected by the load sensor, and to correct an error due to at least one of a movement or a tilt of the movable body.
(12) In this fuel remaining amount measuring apparatus, the detection of the liquid level in the fuel tank with a float is not utilized, but the fuel remaining amount measurement in which an error due to the movement or tilt of the movable body is corrected is performed, which improves the accuracy of the fuel remaining amount measurement.
(13) Preferred embodiments of the present invention are realized in various aspects as fuel remaining amount measuring apparatuses, boats or movable bodies including fuel remaining amount measuring apparatuses, fuel remaining amount measuring methods, and the like.
(14) The above and other elements, features, steps, characteristics and advantages of the present invention will become more apparent from the following detailed description of the preferred embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a diagram schematically illustrating a configuration of a boat according to a first preferred embodiment of the present invention.
(2) FIG. **2** is a diagram illustrating a configuration of a fuel remaining amount measuring apparatus according to the first preferred embodiment of the present invention.
(3) FIG. **3** is a flowchart illustrating a fuel remaining amount measurement process performed by

the fuel remaining amount measuring apparatus according to the first preferred embodiment of the present invention.
(4) FIG. **4** is a diagram illustrating a configuration of a fuel remaining amount measuring apparatus according to a second preferred embodiment of the present invention.
(5) FIG. **5** is a flowchart illustrating a fuel remaining amount measurement process performed by the fuel remaining amount measuring apparatus according to the second preferred embodiment of the present invention.
(6) FIG. **6** is a diagram illustrating a configuration of a fuel remaining amount measuring apparatus according to a third preferred embodiment of the present invention.
(7) FIG. **7** is a flowchart illustrating a fuel remaining amount measurement process performed by the fuel remaining amount measuring apparatus according to the third preferred embodiment of the present invention.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

First Preferred Embodiment

(8) FIG. **1** is a diagram schematically illustrating a configuration of a boat **100** in a first preferred embodiment of the present invention. In FIG. **1** and other figures mentioned below, arrows represent directions defined based on a position of the boat **100**. To be more specific, in each of the figures, arrows represent a forward direction (FRONT), a rearward direction (REAR), an upward direction (UPPER), and a downward direction (LOWER), respectively.
(9) The boat **100** includes a hull **50**, a propulsion device **60**, and a fuel remaining amount measuring apparatus **10**. The hull **50** is a section of the boat **100** where a crew is on board.
(10) The propulsion device **60** generates a propulsion force to propel the boat **100**. In the present preferred embodiment, the propulsion device **60** is at least partially located in a lower space in the hull **50**. The propulsion device **60** may be located in a different place (behind the hull **50**, for instance). The propulsion device includes a power source **62** including an internal combustion engine, for instance, and a propulsion force generation mechanism **64** such as a propeller that is driven by a driving force from the power source **62** so as to generate the propulsion force. The propulsion device may include a steering mechanism and a shift mechanism, neither of which is illustrated. The steering mechanism steers the boat **100**. The shift mechanism is operable to change between a headway state where the driving force from the power source **62** is transmitted to the propulsion force generation mechanism **64** in a direction enabling the boat **100** to make headway, a sternway state where the driving force from the power source **62** is transmitted to the propulsion force generation mechanism **64** in a direction enabling the boat **100** to make sternway, and a neutral state where the driving force from the power source **62** is not transmitted to the propulsion force generation mechanism **64**.
(11) The fuel remaining amount measuring apparatus **10** measures a remaining amount of fuel on the boat **100**. FIG. **2** is a diagram illustrating a configuration of the fuel remaining amount measuring apparatus **10** according to the first preferred embodiment. As illustrated in FIGS. **1** and **2**, the fuel remaining amount measuring apparatus **10** includes a fuel tank **20**, a load sensor **30**, and a correction device **40**.
(12) The fuel tank **20** contains fuel F (diesel oil or gasoline, for instance) to be fed to the power source **62** of the propulsion device **60**. In the present preferred embodiment, the fuel tank **20** is located in the lower space in the hull **50**. As illustrated in FIG. **2**, the fuel tank **20** has a horizontal cross-sectional area varying according to a position in a vertical direction. To be more specific, the fuel tank **20** includes an upper portion that is substantially rectangular-parallelepiped in shape and a lower portion that is narrowed so that the horizontal cross-sectional area may be smaller at a lower position. As a result, the amount of lowering of a liquid level of the fuel F per unit volume of the decreasing fuel F is uniform when the liquid level is in the upper portion of the fuel tank **20**, and is sequentially increased as the liquid level lowers when the liquid level is in the lower portion of the fuel tank **20**. Thus, the fuel tank **20** in the present preferred embodiment has a shape that may cause

a reduction in the measurement accuracy if the fuel remaining amount measuring method, in which a float is used to detect the liquid level in the fuel tank **20** so as to measure the remaining amount of fuel, is used.
(13) The load sensor **30** measures the weight of the fuel tank **20**, and includes a load cell, for instance. In the present disclosure, the weight of the fuel tank **20** refers to a weight of the fuel tank **20** including the weight of the fuel F contained in the fuel tank **20**. In other words, the weight of the fuel tank **20** varies with the remaining amount of the fuel F in the fuel tank **20**. In the present preferred embodiment, the fuel remaining amount measuring apparatus **10** includes n (n being 2 or a greater integer) load sensors **30** (**30**(**1**), **30**(**2**), . . . , and **30**(*n*)). The load sensors **30** support the fuel tank **20** from below the fuel tank **20** and as such output measured load values (LC(1), LC(2), . . . , and LC(n)) corresponding to the weight of the fuel tank **20**. A measured value T1 of the weight of the fuel tank **20** detected by the load sensors is the sum of values LC output from the respective load sensors **30**, as expressed as Formula (1) below.
$$T1=LC(1)+LC(2)+\ldots+LC(n) \qquad (1)$$
(14) The correction device **40** receives the measured value T1 of the weight of the fuel tank **20** detected by the load sensors **30**, and corrects an error due to at least one of a movement or a tilt of the boat **100**. To be specific, the measured value T1 of the weight of the fuel tank **20** detected by the load sensors **30** may include an error due to at least one of the movement and the tilt of the boat **100**. The correction device **40** corrects such error so as to calculate weight TO of the fuel tank **20** in a reference state (a state where the speed and the tilt of the boat **100** are both zero). The correction device **40** includes an acceleration sensor **41**, a calculation unit **42**, and a storage unit **43**.
(15) The acceleration sensor **41** detects acceleration of the boat **100**, and is fitted to the hull **50**. In the present preferred embodiment, the acceleration sensor **41** detects the acceleration in a plurality of directions. Specifically, the acceleration sensor **41** detects and outputs accelerations (gx, gy, and gz) in directions where three axes (x-axis, y-axis, and z-axis) orthogonal to one another extend, respectively.
(16) The calculation unit **42** performs an operation for error correction and includes a central processing unit (CPU), a multi-core CPU or a programmable device (such as a field-programmable gate array (FPGA) and a programmable logic device (PLD)), for instance. The storage unit **43** includes a read-only memory (ROM), a random access memory (RAM), a hard disk drive (HDD), or a solid-state drive (SSD), for instance. The storage unit **43** stores various kinds of programs and data, and is used as a work area and a data storage area required when various processes are performed. As an example, a value of weight Te in the reference state of the fuel tank **20**, which does not contain the fuel F (that is to say, in which the remaining amount of the fuel F is zero), values (gx0, gy0, and gz0) output from the acceleration sensor **41** in the reference state for the respective axes, and specific gravity d of the fuel F contained in the fuel tank **20** are stored in the storage unit **43**.
(17) FIG. **3** is a flowchart illustrating a fuel remaining amount measurement process performed by the fuel remaining amount measuring apparatus **10** according to the first preferred embodiment. The fuel remaining amount measurement process measures the remaining amount of the fuel F in the fuel tank **20**.
(18) Initially, the calculation unit **42** of the correction device **40** acquires the measured value T1 of the weight of the fuel tank **20** detected by the load sensors **30** (step S**110**). As described above, the measured value T1 of the weight of the fuel tank **20** detected by the load sensors **30** may include an error due to at least one of the movement and the tilt of the boat **100**. The calculation unit **42** also acquires acceleration values gx, gy, and gz detected by the acceleration sensor **41** (step S**120**).
(19) Next, the calculation unit **42** of the correction device **40** corrects the error based on the values detected by the acceleration sensor **41** so as to calculate the weight TO of the fuel tank **20** in the reference state (step S**130**). If correction is to be made with respect to the acceleration in the direction of the y-axis in FIG. **2** (acceleration caused by tilt about a longitudinal direction), for

instance, the calculation unit **42** calculates the weight TO of the fuel tank **20** in the reference state according to Formula (2) below. In the formula, gy represents a current acceleration value in the direction of the y-axis output by the acceleration sensor **41**, and gy0 represents the acceleration value in the direction of the y-axis output by the acceleration sensor **41** in the reference state and stored in the storage unit **43**.

$$T0=T1\times(gy0/gy) \qquad (2)$$

(20) Next, the calculation unit **42** of the correction device **40** calculates the weight (weight in the reference state) E0 of the fuel F remaining in the fuel tank **20** according to Formula (3) below (step S**140**). In the formula, Te represents the weight (weight in the reference state) of the fuel tank **20** not containing the fuel F as stored in the storage unit **43**.

$$E0=T0-Te \qquad (3)$$

(21) Next, the calculation unit **42** of the correction device **40** calculates a volume V0 of the fuel F remaining in the fuel tank **20** according to Formula (4) below (step S**150**). In the formula, d represents the specific gravity of the fuel F stored in the storage unit **43**.

$$V0=E0/d \qquad (4)$$

(22) Based on the remaining amount (the volume V0) of the fuel F in the fuel tank **20** thus calculated, the calculation unit **42** of the correction device **40** carries out a fuel remaining amount display through a display device not illustrated (step S**160**).

(23) In the present preferred embodiment, where the fuel remaining amount measurement is performed by the fuel remaining amount measuring apparatus **10** as described above, the detection of the liquid level in the fuel tank with a float is not utilized, but the fuel remaining amount measurement in which an error due to the movement or tilt of the boat **100** is corrected is performed, which improves the accuracy of the fuel remaining amount measurement.

Second Preferred Embodiment

(24) FIG. **4** is a diagram illustrating a configuration of a fuel remaining amount measuring apparatus **10***a* according to a second preferred embodiment. In the description below, components of the fuel remaining amount measuring apparatus **10***a* in the second preferred embodiment that are identical to the components of the fuel remaining amount measuring apparatus **10** in the first preferred embodiment are given identical reference signs, so description thereof will be omitted as appropriate.

(25) The fuel remaining amount measuring apparatus according to the second preferred embodiment includes a dummy weight **22** and a dummy load sensor **32**. The dummy weight **22** is a weight M0 in the reference state that is known. The weight M0 in the reference state of the dummy weight **22** is stored in the storage unit **43** of the correction device **40**. The dummy load sensor **32** measures the weight M1 of the dummy weight **22**, and includes a load cell, for instance. The dummy load sensor **32** supports the dummy weight **22** from below, and as such outputs a measured load value LC(s) corresponding to the weight M1 of the dummy weight **22**. In the present preferred embodiment, the fuel remaining amount measuring apparatus **10***a* includes a single dummy load sensor **32**, so that a measured value of the weight M1 of the dummy weight **22** detected by the dummy load sensor **32** is equal to a value LC(s) output from the dummy load sensor **32**. The fuel remaining amount measuring apparatus **10***a* may include a plurality of dummy load sensors **32** so as to measure the weight M1 of the dummy weight **22** with those dummy load sensors **32**. In the second preferred embodiment, the correction device **40** does not include the acceleration sensor **41**.

(26) FIG. **5** is a flowchart illustrating a fuel remaining amount measurement process performed by the fuel remaining amount measuring apparatus **10***a* in the second preferred embodiment. In the description below, steps of the fuel remaining amount measurement process in the second preferred embodiment that are identical in content to the steps of the fuel remaining amount measurement process in the first preferred embodiment are given identical reference signs, so description thereof will be omitted as appropriate.

(27) In the fuel remaining amount measurement process in the second preferred embodiment, the

calculation unit **42** of the correction device **40** performs a process to acquire the measured value of the weight M1 of the dummy weight **22** detected by the dummy load sensor **32** (step S**122**), instead of a process to acquire the acceleration values detected by the acceleration sensor **41** (step S**120** in FIG. **3**).

(28) Next, the calculation unit **42** of the correction device **40** corrects the error based on the weight M0 in the reference state of the dummy weight **22** as stored in the storage unit **43** and the measured value of the weight M1 of the dummy weight **22** detected by the dummy load sensor **32**, so as to calculate the weight TO of the fuel tank **20** in the reference state (step S**132**). Specifically, the calculation unit **42** calculates the weight TO of the fuel tank **20** in the reference state according to Formula (5) below. The subsequent process steps (steps S**140** through S**160**) are identical to the process steps in the fuel remaining amount measurement process in the first preferred embodiment.

$T0=T1\times(M0/M1)$ (5)

(29) Also in the second preferred embodiment, where the fuel remaining amount measurement is performed by the fuel remaining amount measuring apparatus **10***a* as described above, the detection of the liquid level in the fuel tank **20** with a float is not utilized, but the fuel remaining amount measurement in which an error due to the movement or tilt of the boat **100** is corrected is performed, which improves the accuracy of the fuel remaining amount measurement.

Third Preferred Embodiment

(30) FIG. **6** is a diagram illustrating a configuration of a fuel remaining amount measuring apparatus **10***b* according to a third preferred embodiment. In the description below, components of the fuel remaining amount measuring apparatus **10***b* in the third preferred embodiment that are identical to the components of the fuel remaining amount measuring apparatus **10** in the first preferred embodiment are given identical reference signs, so description thereof will be omitted as appropriate.

(31) In the fuel remaining amount measuring apparatus **10***b* according to the third preferred embodiment, the correction device **40** includes a tilt sensor **45** instead of the acceleration sensor **41**. The tilt sensor **45** detects the tilt of the boat **100**, and is fitted to the hull **50**. In the storage unit **43** of the correction device **40**, a value of a correction coefficient α corresponding to an output from the tilt sensor **45** (namely, the tilt of the boat **100**), which coefficient is set in advance, is stored.

(32) FIG. **7** is a flowchart illustrating a fuel remaining amount measurement process performed by the fuel remaining amount measuring apparatus **10***b* according to the third preferred embodiment. In the description below, steps of the fuel remaining amount measurement process in the third preferred embodiment that are identical in content to the steps of the fuel remaining amount measurement process in the first preferred embodiment are given identical reference signs, so description thereof will be omitted as appropriate.

(33) In the fuel remaining amount measurement process according to the third preferred embodiment, the calculation unit **42** of the correction device **40** performs a process to acquire a value of the tilt of the boat **100** detected by the tilt sensor **45** (step S**124**), instead of the process to acquire the acceleration values detected by the acceleration sensor **41** (step S**120** in FIG. **3**).

(34) Next, the calculation unit **42** of the correction device **40** corrects the error based on the value of the tilt detected by the tilt sensor **45** so as to calculate the weight TO of the fuel tank **20** in the reference state (step S**134**). Specifically, the calculation unit **42** calculates the weight TO of the fuel tank **20** in the reference state according to Formula (6) below. In the formula, a represents a correction coefficient corresponding to the output from the tilt sensor **45**, as set in advance and stored in the storage unit **43**. The subsequent process steps (steps S**140** through S**160**) are identical to the process steps in the fuel remaining amount measurement process in the first preferred embodiment.

$T0=T1\times\alpha$ (6)

(35) Also in the third preferred embodiment, where the fuel remaining amount measurement is performed by the fuel remaining amount measuring apparatus **10***b* as described above, the detection

of the liquid level in the fuel tank **20** with a float is not utilized, but the fuel remaining amount measurement in which an error due to the movement or tilt of the boat **100** is corrected is performed, which improves the accuracy of the fuel remaining amount measurement.
(36) Modifications
(37) The present invention is not limited to the preferred embodiments described above, but may be modified in various ways without departing from the gist of the present invention. As an example, such modifications described below are possible.
(38) The configurations of the boat **100** and the fuel remaining amount measuring apparatuses **10**, **10***a*, and **10***b* in the above preferred embodiments are presented as mere examples, and are variously modifiable. For instance, the fuel remaining amount measuring apparatuses **10**, **10***a*, and **10***b* in the above preferred embodiments each include a plurality of load sensors **30**, although the fuel remaining amount measuring apparatuses **10**, **10***a*, and **10***b* may each include a single load sensor **30**. In the first preferred embodiment, the acceleration sensor **41** detects the accelerations in the directions, where the three axes (x-axis, y-axis, and z-axis) are orthogonal to one another, respectively. The acceleration sensor **41** may detect acceleration in one or more optional directions.
(39) The contents of the fuel remaining amount measurement process in the above preferred embodiments are presented as mere examples, and are variously modifiable. For instance, in the above preferred embodiments, the remaining amount of fuel is finally calculated as the volume so as to provide the fuel remaining amount display, although the remaining amount of fuel may be calculated as the weight so as to provide the fuel remaining amount display based on the calculated weight.
(40) In the above preferred embodiments, the fuel tank **20** is structured with a horizontal cross-sectional area varying according to the position in the vertical direction, although the technology disclosed in the present invention is equally applicable to the fuel remaining amount measurement in a fuel tank having another shape.
(41) In the above preferred embodiments, the fuel remaining amount measurement in the fuel tank **20** of the boat **100** is described, although the technology disclosed in the present disclosure is equally applicable to the fuel remaining amount measurement in a fuel tank of a movable body such as a vehicle or an aircraft.
(42) While preferred embodiments of the present invention have been described above, it is to be understood that variations and modifications will be apparent to those skilled in the art without departing from the scope and spirit of the present invention. The scope of the present invention, therefore, is to be determined solely by the following claims.

Claims

1. A fuel remaining amount measuring apparatus for a boat, the fuel remaining amount measuring apparatus comprising: a fuel tank containing fuel; a load sensor to measure a weight of the fuel tank; and a corrector to receive a measured value T1 of the weight of the fuel tank detected by the load sensor, and to correct an error due to at least one of a movement or a tilt of the boat; wherein the corrector includes: an acceleration sensor to detect an acceleration of the boat; a calculator to correct the error based on a value g of the acceleration detected by the acceleration sensor; and a storage to store a value g0 output from the acceleration sensor in a reference state where a speed and a tilt of the boat are both zero; and the calculator corrects the error by multiplying the measured value T1 of the weight of the fuel tank detected by the load sensor by a value g0/g.
2. The fuel remaining amount measuring apparatus according to claim 1, wherein the acceleration sensor is operable to detect at least an acceleration caused by the tilt of the boat about a longitudinal direction of the boat.
3. The fuel remaining amount measuring apparatus according to claim 1, wherein the acceleration sensor is operable to detect the acceleration of the boat in a plurality of directions.

4. The fuel remaining amount measuring apparatus according to claim 1, wherein the corrector includes: a tilt sensor to detect the tilt of the boat; and the calculator corrects the error based on a value of the tilt detected by the tilt sensor.
5. The fuel remaining amount measuring apparatus according to claim 1, wherein the storage stores a data value reflective of the weight of the fuel tank in a state of not containing fuel.
6. The fuel remaining amount measuring apparatus according to claim 5, wherein the storage is operable to store a specific gravity of the fuel.
7. The fuel remaining amount measuring apparatus according to claim 1, wherein the fuel remaining amount measuring apparatus comprising a plurality of load sensors, and the load sensor is one of the plurality of load sensors.
8. The fuel remaining amount measuring apparatus according to claim 1, wherein a structure of the fuel tank has a horizontal cross-sectional area that varies according to a position in a vertical direction.
9. A boat comprising: a hull; a propulsion device fitted to the hull; and the fuel remaining amount measuring apparatus according to claim 1; wherein the propulsion device is operable to use the fuel contained in the fuel tank to generate a propulsion force.
10. A fuel remaining amount measuring apparatus for a boat, the fuel remaining amount measuring apparatus comprising: a fuel tank containing fuel; a load sensor to measure a weight of the fuel tank; a corrector to receive a measured value of the weight of the fuel tank detected by the load sensor, and to correct an error due to at least one of a movement or a tilt of the boat; a dummy weight; a storage to store a weight of the dummy weight; a dummy load sensor to measure the weight of the dummy weight; and a calculator to correct the error based on the weight of the dummy weight stored in the storage and a measured value of the weight of the dummy weight detected by the dummy load sensor.
11. A boat comprising: a hull; a propulsion device fitted to the hull; and the fuel remaining amount measuring apparatus according to claim 5; wherein the propulsion device is operable to use the fuel contained in the fuel tank to generate a propulsion force.
12. A fuel remaining amount measuring apparatus for a movable body, the fuel remaining amount measuring apparatus comprising: a fuel tank containing fuel; a load sensor to measure a weight of the fuel tank; and a corrector to receive a measured value T1 of the weight of the fuel tank detected by the load sensor, and to correct an error due to at least one of a movement or a tilt of the movable body; wherein the corrector includes: an acceleration sensor to detect an acceleration of the movable body; a calculator to correct the error based on a value g of the acceleration detected by the acceleration sensor; and a storage to store a value g0 output from the acceleration sensor in a reference state where a speed and a tilt of the movable body are both zero; and the calculator corrects the error by multiplying the measured value T1 of the weight of the fuel tank detected by the load sensor by the value g0/g.
13. A movable body comprising: a main body; a propulsion device fitted to the main body; and the fuel remaining amount measuring apparatus according to claim 12; wherein the propulsion device is operable to use the fuel contained in the fuel tank to generate a propulsion force.
14. A fuel remaining amount measuring apparatus for a movable body, the fuel remaining amount measuring apparatus comprising: a fuel tank containing fuel; a load sensor to measure a weight of the fuel tank; a corrector to receive a measured value of the weight of the fuel tank detected by the load sensor, and to correct an error due to at least one of a movement or a tilt of the movable body; a dummy weight; a storage to store a weight of the dummy weight; a dummy load sensor to measure the weight of the dummy weight; and a calculator to correct the error based on the weight of the dummy weight stored in the storage and a measured value of the weight of the dummy weight detected by the dummy load sensor.
15. A movable body comprising: a main body; a propulsion device fitted to the main body; and the

fuel remaining amount measuring apparatus according to claim 14; wherein the propulsion device is operable to use the fuel contained in the fuel tank to generate a propulsion force.